

W019 Strain Build List

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Forty-five strains to construct: 25 double knockouts and 20 triple knockouts. Every one of them is made from a strain that already exists, so no new single knockout is needed and no triple waits on one of the 25 new doubles being finished. The new doubles are needed for the measurement rather than for the construction: τ on a triple is computed from all three of its pairs and all three of its singles, so a triple measured without them yields a fitness and no interaction. Tables 3 and 4 are the build list. Tables 1 and 2 are the strains in hand, and Table 5 is the plate they are all measured on.

Tables generated by `experiments/W019-echo-crispr-array/scripts/build_list_tables.py`

Working notes: `notes/experiments.W019-echo-crispr-array/*.md`

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1 Terms ✗

boot SE The standard error of a strain's mean fitness: resample the units the mean was taken over, take the mean of each resample, and report the spread of those means. Both *boot SE* columns of Table 1 are this same statistic, and what differs is the unit resampled. Ours resamples the three replicate plates. Costanzo's resamples the seventeen replicate control screens standing behind a query-strain value, with the colonies of a screen averaged before the resampling, so their unit is a whole screen and there are seventeen of them against our three. Two consequences worth stating: their number is the tighter one for the same underlying spread, and a strain-by-strain comparison of the two is therefore not like for like. Costanzo publishes the column under the name standard deviation. It is the spread of bootstrapped means, which makes it a standard error, and it enters our loaders as one, typed `bootstrap_se`.

colony SD The published uncertainty on a double (Table 2), and a different kind of number from either *boot SE*: a sample standard deviation over the four to eight replicate colonies within one screen, which the authors warn underestimates the true variance because adjacent colonies are not independent. Being a spread rather than an error on a mean, it compares to our *plate SD*. Costanzo publishes no bootstrap standard error for a double.

fitness Colony growth relative to the wild type on the same plate, so 1 is wild-type growth.

plate SD The standard deviation of a strain's fitness across the replicate plates it was measured on. It is larger than that strain's *boot SE*, which is the error on the mean of those same plates.

routes How many different ways the bench can make a triple from strains that already exist. Every triple is built by crossing one existing double with the remaining single. *routes* 1 means only one of the triple's three pairs is in the freezer, so there is a single way in, and a failed cross blocks that triple until the matching new double from Table 3 is done. *routes* 2 means two of its pairs exist, so a failed cross has an immediate second path.

tier How a double was classed when run 4 was designed. **validation** reproduces a measurement someone has already published, **coverage** was added so that later triples would have a parent to be built from, and **novel** had not been measured by anyone.

2 Strains that already exist ✗

Twelve singles and thirteen doubles were built and measured in run 4. No triple has been constructed in any round.

Table 1: Single knockouts that already exist, all twelve measured in run 4. The two *boot SE* columns are the same statistic, the spread of a bootstrapped mean, and differ only in what is resampled: our three replicate plates against the seventeen control screens behind a query-strain published value, whose colonies are averaged within a screen first. Costanzo publishes that column under the name standard deviation; it is a standard error and is used as one here. *plate SD* is the other kind of number, the spread across our plates rather than the error on their mean, and the singles table has no published counterpart to it. Sec. 1 carries the comparison.

id	ORF	common	Ours, run 4			Costanzo 2016	
			fitness	boot SE	plate SD	SMF	boot SE
				over 3 plates			over 17 screens
s1	YJR060W	CBF1	0.5348	0.0506	0.1076	0.5900	0.1138
s2	YPL081W	RPS9A	0.9007	0.0515	0.1105	0.9550	0.1553
s3	YDR057W	YOS9	1.2430	0.1282	0.2690	1.0435	0.0535
s4	YPL046C	ELC1	1.2066	0.0495	0.1052	1.0433	0.0643
s5	YBR203W	COS111	1.1292	0.0679	0.1426	1.0370	0.0458
s6	YGL087C	MMS2	1.1628	0.0329	0.0706	0.9960	0.0940
s7	YLR104W	LCL2	1.1748	0.0564	0.1187	1.0322	0.0453
s8	YLL012W	YEH1	1.2604	0.0659	0.1402	0.9925	0.0804
s9	YER079W	–	1.3333	0.1504	0.3162	1.0387	0.1173
s10	YKL033W-A	–	0.9251	0.0445	0.0954	1.0327	0.1008
s11	YLR313C	SPH1	1.0882	0.1471	0.3083	0.9843	0.0271
s12	YLR312C-B	–	0.8390	0.0661	0.1397	1.0845	0.0441

Eleven of the twelve singles carry a prediction, not twelve. The panel the interaction predictions were made over holds *YIL174W*, which has not been built, and does not hold SPH1 (*YLR313C*), which has. So of the twelve singles in Table 1, eleven are panel genes with a prediction attached and one is not, and every triple in this build list is drawn from those eleven.

One of the eleven is itself unsettled. *YLR312C-B* (s12) is an SGD *ORF*, *Merged* feature with no discrete protein, it carries zero trigenic training records, and SPH1 was the recommended replacement for it, a swap that has not been made. It is in six of the twenty triples. Setting it aside would leave ten usable panel genes and would change those six, so the decision is worth making before the crosses start rather than after.

SPH1 is an open question. s11, SPH1 (*YLR313C*), is in none of the twenty triples in Table 4, at a measured fitness of 1.0882 (Table 1). It can ride along on the measurement plate as a near-wild-type control, or it can be dropped. If it rides along, the plate carries 71 strains rather than 70, at the same five wells each.

Table 2: Double knockouts that already exist, all thirteen measured in run 4. Column attribution is as in Table 1. *colony SD* is the published spread over the four to eight replicate colonies of one screen, which the authors warn underestimates because adjacent colonies are not independent. It is a spread and not an error on a mean, so it compares to our *plate SD* and not to either *boot SE*; Costanzo publishes no bootstrap SE for a double. *parent* says whether the pair is the starting strain for one of the triples in Table 4. A dash in the Costanzo columns means no one has published the pair.

id	pair	Ours, run 4			Costanzo 2016			
		fitness	boot SE	plate SD	DMF	colony SD	tier	parent
d1	YDR057W + YPL081W	0.6071	0.0479	0.1029	1.0351	0.0447	coverage	yes
d2	YPL046C + YPL081W	0.9335	0.0040	0.0084	–	–	novel	no
d3	YER079W + YPL081W	0.9496	0.0556	0.1195	0.8625	0.0975	validation	no
d4	YLR312C-B + YPL081W	0.6480	0.0454	0.0968	1.0165	0.0418	coverage	yes
d5	YDR057W + YGL087C	0.7884	0.0007	0.0015	1.1372	0.0238	validation	yes
d6	YDR057W + YER079W	0.9403	0.0472	0.0991	1.0963	0.0541	coverage	no
d7	YDR057W + YLR312C-B	0.8895	0.0543	0.1150	1.1783	0.0257	validation	no
d8	YJR060W + YPL046C	0.5649	0.0435	0.0930	0.9663	0.0178	coverage	yes
d9	YBR203W + YPL046C	0.8882	0.0232	0.0501	1.1179	0.0171	coverage	yes
d10	YDR057W + YLL012W	0.9315	0.0269	0.0572	0.9995	0.0690	coverage	yes
d11	YLL012W + YPL046C	0.9087	0.0348	0.0744	1.0251	0.0534	coverage	yes
d12	YER079W + YJR060W	0.6886	0.0650	0.1365	0.6100	0.0183	validation	no
d13	YER079W + YLR312C-B	0.7728	0.0514	0.1084	1.0994	0.0393	coverage	yes

All thirteen doubles go on the measurement plate. The pairs marked **yes** under *parent* are the starting strains for Table 4; the rest are re-measured alongside them, which gives a second reading on each of those pairs.

One designed double failed to construct: no colonies after transformation. $YJR060W \times YKL033W-A$ was designed into run 4 and is the one pair of the fourteen that did not come through: the transformation gave zero colonies. No attempt date was recorded and no retry is on record. Both parent singles exist and were measured. The pair is not part of the build list, and nothing in Tables 3 or 4 depends on it.

3 The 25 doubles to construct ✗

Cross the two singles named in the row. Both parents are already in the freezer (Table 1) and no row depends on another, so all 25 can be started at once. Each row gives the ORF and, where the gene has one, the common name in parentheses.

Table 3: The 25 double knockouts to construct, one per row. Each row names the two singles to cross, both of them in Table 1.

ID	gene 1	gene 2
D01	YBR203W (COS111)	YDR057W (YOS9)
D02	YBR203W (COS111)	YGL087C (MMS2)
D03	YBR203W (COS111)	YJR060W (CBF1)
D04	YBR203W (COS111)	YKL033W-A
D05	YBR203W (COS111)	YLL012W (YEH1)
D06	YBR203W (COS111)	YLR104W (LCL2)
D07	YBR203W (COS111)	YLR312C-B
D08	YBR203W (COS111)	YPL081W (RPS9A)
D09	YDR057W (YOS9)	YKL033W-A
D10	YER079W	YLL012W (YEH1)
D11	YER079W	YLR104W (LCL2)
D12	YGL087C (MMS2)	YLL012W (YEH1)
D13	YGL087C (MMS2)	YLR312C-B
D14	YGL087C (MMS2)	YPL046C (ELC1)
D15	YGL087C (MMS2)	YPL081W (RPS9A)
D16	YJR060W (CBF1)	YLL012W (YEH1)
D17	YJR060W (CBF1)	YLR104W (LCL2)
D18	YKL033W-A	YLL012W (YEH1)
D19	YKL033W-A	YLR312C-B
D20	YKL033W-A	YPL046C (ELC1)
D21	YKL033W-A	YPL081W (RPS9A)
D22	YLL012W (YEH1)	YLR312C-B
D23	YLR104W (LCL2)	YLR312C-B
D24	YLR104W (LCL2)	YPL046C (ELC1)
D25	YLR104W (LCL2)	YPL081W (RPS9A)

4 The 20 triples to construct ✗

Cross the existing double in *build from* with the single in *third single*. Start with T01 through T15, which have one route each: a failed cross there blocks the triple until the matching double from Table 3 is finished, so those failures are worth surfacing early. T16 through T20 each have a second route already in hand and can absorb a failed cross without waiting on anything.

Table 4: The 20 triple knockouts to construct, in build order. *build from* names an existing double by ORF and *third single* the remaining gene; crossing the two gives the triple. *routes* counts how many different ways the bench can make the triple from strains that already exist, and is defined in Sec. 1.

ID	gene 1	gene 2	gene 3	build from	× third single	routes
T01	YKL033W-A	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YKL033W-A	1
T02	YER079W	YLL012W (YEH1)	YLR312C-B	YER079W + YLR312C-B	YLL012W (YEH1)	1
T03	YLR104W (LCL2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YLR104W (LCL2)	1
T04	YBR203W (COS111)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YBR203W (COS111)	1
T05	YER079W	YLR104W (LCL2)	YLR312C-B	YER079W + YLR312C-B	YLR104W (LCL2)	1
T06	YGL087C (MMS2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YGL087C (MMS2)	1
T07	YBR203W (COS111)	YDR057W (YOS9)	YPL081W (RPS9A)	YDR057W + YPL081W	YBR203W (COS111)	1
T08	YDR057W (YOS9)	YKL033W-A	YPL081W (RPS9A)	YDR057W + YPL081W	YKL033W-A	1
T09	YDR057W (YOS9)	YKL033W-A	YLL012W (YEH1)	YDR057W + YLL012W	YKL033W-A	1
T10	YGL087C (MMS2)	YLL012W (YEH1)	YPL046C (ELC1)	YLL012W + YPL046C	YGL087C (MMS2)	1
T11	YBR203W (COS111)	YLR104W (LCL2)	YPL046C (ELC1)	YBR203W + YPL046C	YLR104W (LCL2)	1
T12	YBR203W (COS111)	YKL033W-A	YPL046C (ELC1)	YBR203W + YPL046C	YKL033W-A	1
T13	YJR060W (CBF1)	YLR104W (LCL2)	YPL046C (ELC1)	YJR060W + YPL046C	YLR104W (LCL2)	1
T14	YBR203W (COS111)	YGL087C (MMS2)	YPL046C (ELC1)	YBR203W + YPL046C	YGL087C (MMS2)	1
T15	YBR203W (COS111)	YDR057W (YOS9)	YLL012W (YEH1)	YDR057W + YLL012W	YBR203W (COS111)	1
T16	YDR057W (YOS9)	YGL087C (MMS2)	YPL081W (RPS9A)	YDR057W + YGL087C	YPL081W (RPS9A)	2
T17	YJR060W (CBF1)	YLL012W (YEH1)	YPL046C (ELC1)	YJR060W + YPL046C	YLL012W (YEH1)	2
T18	YBR203W (COS111)	YLL012W (YEH1)	YPL046C (ELC1)	YBR203W + YPL046C	YLL012W (YEH1)	2
T19	YBR203W (COS111)	YJR060W (CBF1)	YPL046C (ELC1)	YBR203W + YPL046C	YJR060W (CBF1)	2
T20	YDR057W (YOS9)	YGL087C (MMS2)	YLL012W (YEH1)	YDR057W + YGL087C	YLL012W (YEH1)	2

5 The measurement plate ✗

Seventy strains on one 384-well layout, five wells each, one pick per strain.

A work in progress. It gives the shape of the assay, not the layout that will be run, and the counts are approximate: they fix how many strains the plate carries, not where any of them goes. The twenty wild-type wells are a working figure, thirteen wells are spare, and SPH1 can ride along as a seventy-first strain without changing the five each. Every strain fits on one plate and several plates can be run; how many biological replicates to spread across them is the open question, since each plate is scored against its own wild-type wells and a plate whose wild type sits low shifts every fitness on it, as run-3 P2 did. The final well assignment comes from the picklist.

Table 5: What goes on the measurement plate, approximately. 70 strains at 5 wells each, one 384-well layout, one pick per strain. Wells per strain is the whole number of times 70 strains divide the 358 wells left after 6 are kept empty to fix the plate’s orientation and 20 are given to the wild-type reference, which gives 5. The 20 wild-type wells are a working figure rather than a fixed allocation, and 13 wells stay spare, so these are strain counts and not a well assignment; the picklist decides that. Every existing double is re-measured alongside the new strains, which gives a second reading on each of those pairs.

on the plate	strains	wells
singles, already built	11	55
doubles, already built (Table 2)	13	65
doubles, to construct (Table 3)	25	125
triples, to construct (Table 4)	20	100
wild type, BY4741	1	20
total	70	365
orientation blanks	–	6
spare	–	13
plate		384